

Scalability & Future Plan

Date	29-09-2026
Team ID	SWTID-2026-5359
Project Name	Courier Management System Using Salesforce
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current Courier Management System Using Salesforce solution can be scaled to handle larger user bases (10k+ customers, 1000+ delivery agents), increased shipment data loads (1M+ records), or extended features like AI route optimization and multi-country logistics in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High/Medium/Low)
1	Manual agent assignment for shipments causes delays	Delivery delays, inefficient agent utilization, customer dissatisfaction	High
2	Limited real-time tracking and no GPS integration for delivery agents	Lack of visibility for customers, difficult to track delivery status	High
3	Single org storage limitation for large shipment history (100k+ records)	Performance degradation, slow reports, storage limit exceeded	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	100-500 users (customers + agents) in single Salesforce org	Implement Salesforce Platform Events, Queueable Apex for async processing, Multi-org architecture with MuleSoft integration to handle 10,000+ users
2	Data Storage	Custom Objects with 100k records, limited file storage	Implement Salesforce Big Objects for shipment history archiving (1M+ records), External Objects, Salesforce Files + AWS S3 integration for documents
3	Performance	Synchronous Apex, slow for bulk shipment creation	Use Asynchronous Apex (Batch, Queueable), Platform Cache for pricing engine, CDN for customer portal, Optimized SOQL with indexing
4	Security		Implement Shield Platform Encryption, MFA for all users, Field-Level Security, Health Check, Session Management, 2FA for agents

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	AI-powered route optimization using Google Maps + Einstein AI, Automated agent assignment logic	Q1 2027	30% faster delivery, reduced fuel cost, improved efficiency

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Mobile App enhancement with offline capability for delivery agents, Real-time push notifications via Platform Events, Barcode scanning for shipment verification	Q2 2027	Agents can work offline in low network areas, 50% reduction in missed deliveries, improved tracking accuracy
Phase 3	Multi-country logistics support, Multi-currency payment integration (Razorpay + PayPal), AI Chatbot for customer support using Einstein Bot, Blockchain for shipment verification	Q3 2027	Global courier operations, 24/7 customer support, secure shipment tracking
Phase 4	Predictive analytics for delivery time estimation using Einstein Analytics, Customer loyalty program, Integration with e-commerce platforms (Shopify, WooCommerce), Carbon footprint tracking for green logistics	Q4 2027	Accurate delivery predictions, increased customer retention, eco-friendly logistics, e-commerce growth